

LESSON 1.2

RATIO RELATIONSHIPS



LESSON INTRODUCTION

MA.6.AR.3.1 Given a real-world context, write and interpret ratios to show the relative sizes of two quantities using appropriate notation: $a:b$, a to b , or $a:b$, where $b \neq 0$.

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.

LEARNING TARGETS

- I can interpret ratios when given multiple representations.
- I can solve real-world problems involving ratios.

BENCHMARK COHERENCE

BUILDING ON

MA.5.FR.1.1 Given a mathematical or real-world problem, represent the division of two whole numbers as a fraction.

WORKING TOWARD

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step real-world percent problems.

MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.

ESSENTIAL QUESTIONS

- What are the different ways that ratios can be expressed?
- How can ratios be used to solve problems?
- How can ratios be interpreted?

LESSON NARRATIVE

Students express ratios in several different ways. From a given ratio, they determine unit rates and use the rates to answer real-world questions. They also apply ratios to percentages, representing with models and diagrams, and use them to solve real-world problems.

COMPONENT SUMMARY

1.2.1 WARM-UP (~10 MIN.)

Fill in numbers for missing digits in equations that compare equivalent fractions

1.2.3 GUIDED PRACTICE (5 MIN.)

Represent a ratio relationship in several different ways; apply the ratio given a new “whole”

1.2.5 GUIDED PRACTICE (10 MIN.)

Apply ratios to percentages to solve real-world problems using double number lines and bar models

1.2.2 REFRESHER (5 MIN.)

Review several different ways to express ratios

1.2.4 COLLABORATION (10 MIN.)

Write unit rates from given ratios and use to answer real-world questions

1.2.6 WRAP-UP (~5 MIN.)

Use ratios to solve real-world problems

RESOURCES

GLOSSARY

Key Terms:

- Equivalent fraction
- Ratio
- Percentage

Additional Terms:

- N/A

MATERIALS

- (Optional) Calculators
- (Optional) Colored cubes

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may incorrectly write the stacked ratios in the table, writing the dollar amount on the bottom and the pay rate at the top in 1.2.4 Collaboration.
- Students may struggle with differing between part: part and part: whole ratios.

THINGS TO CONSIDER

- Consider providing a completed bar model for representing percentages for students to reference when working through 1.2.5 Guided Practice and 1.2.6 Wrap-Up.
- Allowing students to use calculators during the lesson will help students focus on recalling the content. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers.
- Students will further utilize their understanding of ratios in unit 2 as they learn about probability. Noting student misconceptions and addressing these will help students be successful in units 2, 5, 6, 7, and 11.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

1.2.1 Warm-Up allows students to engage in solving a question where they are selecting numbers to fill in missing digits in equations that compare equivalent fractions. Consider using the Think-Pair-Share routine to provide students with multiple attempts at making sense, both independently and with a partner for increased precision.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Clarifications in MA.K12.MTR.2.1 prioritize the use of multiple representations to make sense of abstract concepts, including to "... guide students from concrete, to pictorial, to abstract representations..." Utilize the visual models and discussion prompts provided to help students to make this shift.

DIFFERENTIATE INSTRUCTION

This lesson utilizes multiple representations for interpreting ratios. Representations include double number lines, tape diagrams, tables, and visual models. The multiplicity of representations provides a variety of entry points for learners to engage with the content. For students who need a kinesthetic experience to make sense of the concepts, manipulatives can also be incorporated and connected to the visual models provided in the text. Provide opportunities for students to practice flexibility if there are particular models that work best for their learning needs.

1.2.1 WARM-UP: EQUIVALENT FRACTIONS

TEACHER MOVES

Think-Pair-Share

Consider utilizing the Think-Pair-Share strategy here. It may provide a low-stakes opportunity to establish this routine in your classroom. Allow students to independently read and work through the problem, then have them pair up and verify their work with a partner. Bring the class together for a whole-class discussion, sharing approaches that students used when working through the activity. Highlight any student responses that make use of equivalence.



1.2.1 Warm-Up: Equivalent Fractions

- Using the digits 1 to 9 once each, fill the boxes so that the three equations show equivalent fractions.

$$\frac{7}{\boxed{1}2} = \frac{\boxed{2}8}{\boxed{4}\boxed{8}}$$

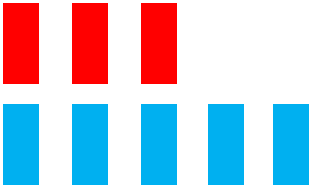
$$\frac{3}{\boxed{5}} = \frac{\boxed{6}}{10}$$

$$\frac{12}{\boxed{3}6} = \frac{2}{6} = \frac{\boxed{9}}{2\boxed{7}}$$



1.2.2 Refresher: Reading and Writing Ratios

1. The Florida Gators are experimenting with creating a new flavor of Gatorade, Gatormax. To create Gatormax, they will mix the following ratio of red and blue Gatorade together.



- a. Complete the statement.
There are 3 red Gatorades for every 5 blue Gatorades in Gatormax.

There are three ways to express ratios.

Method for Writing Ratios	The Ratio of Red Gatorades to Blue Gatorades
Using a colon (:)	3 : 5
Using the word "to"	3 to 5
Using a fraction bar	$\frac{3}{5}$

- b. Write the ratio of blue Gatorades to total Gatorades in three ways.

5 : 8	5 to 8	$\frac{5}{8}$
-------	--------	---------------

1.2.2 REFRESHER: READING AND WRITING RATIOS

TEACHER MOVES

Students use multiple representations to express ratios in question 1b. Encourage them to think about how each relates to the visual model and verbal description provided in question 1a. Prompt them to think about what is being compared and what it means when related to the context of the problem.

- What is being compared? How do you know?
- How many total bottles of Gatorade are represented?
- How many of the bottles are red? Blue?
- For every ___ bottles of Gatorade, ___ are red and ___ are blue.
- When making a batch of Gatormax, what is the minimum number of bottles of Gatorade needed?

1.2.3 GUIDED PRACTICE: REPRESENTING RATIO RELATIONSHIPS

DIFFERENTIATE INSTRUCTION

Students may struggle to transfer the 2:3 relationship to the new total of 25 students presented in part c. Provide red and blue cubes for students and have them model the relationship, creating groups of 2 red and 3 blue cubes to represent each group of 2:3 until 25 cubes are used. When they're finished, they should have 5 groups of 5 total cubes, representing the 25 students in the class.

TEACHER MOVES

When completing the table in part d, students may incorrectly fill in the values as they move across, multiplying each value by 8 since that is the relationship that is given between 3 and 24. Prompt them to look back at the double number line and given ratio in 1a. The values in each row increased differently depending on the value represented in the ratio.

Encourage them to fill in the table by first finding the total number of students (5) and then finding the students who wear glasses when 24 wear no glasses (40). Using the relationship between 40 and 80 should help them to determine the values for students who wear glasses and students who do not.



1.2.3 Guided Practice: Representing Ratio Relationships

1. The ratio of the number of students who wear glasses to the number of students who do not is 2 : 3.

- a. Complete the double number line for this ratio.



- b. If 18 students in the class do not wear glasses, how many students are in the class?

30 students

- c. Use the tape diagram to determine how many students do not wear glasses given that there are 25 students in the class.



15 students

- d. Complete the table.

Glasses	2	16	32
No Glasses	3	24	48
Total Students	5	40	80



1.2.4 Collaboration: Rates and Unit Rates

1. Four students work as baristas after school. They all start out with the same pay per hour. During the first year of employment, the baristas earn a \$0.05 per hour raise in pay each month. Work with your partner to complete the following questions.
- a. Use the relationship between the number of hours worked and the amount of money earned to complete the table.

Employee	Money Earned (\$)	Hours Worked	Rate (\$ per hours)	Unit Rate (\$ per hour)
Elijah	62	4	$\frac{62}{4}$	$\frac{15.5}{1} = 15.50$
Joy	39	2.5	$\frac{39}{2.5}$	$\frac{15.6}{1} = 15.60$
Sharon	173.25	11	$\frac{173.25}{11}$	$\frac{15.75}{1} = 15.75$

- b. Which student has worked as a barista for the longest amount of time?
Sharon
- c. Brian was hired a month after Elijah. What is Brian's rate of pay per hour?
\$15.45
- d. Brian worked 6 hours last week. How much money did Brian earn last week?
\$92.70

1.2.4 COLLABORATION: RATES AND UNIT RATES

TEACHER MOVES

Students may incorrectly write the stacked ratios in the table, writing the dollar amount on the bottom and the pay rate at the top. Remind them that just as they read the ratios from left to write, they should also read a stacked ratio from top to bottom.

ENRICHMENT

Students may struggle to interpret the information in the prompt and table to answer parts b, c, and d. Consider asking questions like:

- How can you use the information in the table to help determine who has worked the longest as a barista?
- What information in the prompt can help you to determine Brian's pay rate? How did you use this information to calculate his rate of pay?

QUESTIONING

- Challenge partners to answer the following questions:
- Which student has worked as a barista for the least amount of time? How do you know?
 - How many months after Sharon was Joy hired? How do you know?
 - How many months after Sharon was Elijah hired? How do you know?
 - Carissa was hired 6 months before Sharon. What is her unit rate of pay?

1.2.5 GUIDED PRACTICE: APPLYING RATIOS TO PERCENTAGES

TEACHER MOVES

Encourage students unfamiliar with a double number line to make connections between the given numbers and correlating percentages. Ask questions like:

- What is it you are trying to find?
- What is the total given? Are 21 and 70 similar in any way? Do they have any common factors? How could you use this information to divide up the first number line into equal intervals?
- How many numbers are represented on the top number line? How can you use this information to determine how to divide up the second number line into equal percentages?

ENRICHMENT

Encourage students to find 10%, as they did when using the tape diagram in question 2. They can then multiply by 6 to find 60% or find 50% and then add 10%.

- Why would it be helpful to find 10% of the original price?
- If you found 50% of the original price, how could you use this to find 60%?



1.2.5 Guided Practice: Applying Ratios to Percentages

A **percentage** is a ratio expressed as a fraction of 100.

Work with your partner to use the diagrams or tables provided to solve the problems.

1. Determine what percentage of 70 is 21. Then, complete the statement.



21 is 30 % of 70.

2. Determine 70% of \$60. Then, complete the statement.

dollars

percentage

70% of \$60 is \$ 42.

3. A pair of jeans is on sale for 60% of its regular price. The regular price of the jeans is \$65.

- a. Complete the ratio table to determine the sale price.

Price (\$)	Percentage
6.5	10
39	60
65	100

- b. Complete the statement.
The sale price of the jeans is \$ 39.



1.2.6 Wrap-Up: Ticket Out the Door

- The table shows the cost of boxes of crayons at a convenience store.
 - Use the ratio table to find the cost of 15 boxes of crayons.

Crayon Boxes	6	15	1
Cost (\$)	9	22.50	1.50

- Find the unit rate in dollars per box.
\$1.50 per box
- Fatima has a 20% off coupon at the store. Her total comes out to \$70. Use a double number line, tape diagram, or ratio table to determine how much her coupon is worth.
\$14.00

1.2.6 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

As this is the first Ticket Out the Door, establishing your routine with students is important. Consider using these to inform your teaching. Keep a log of misconceptions that should be addressed as a whole class or in small groups. This information can also inform your teaching for Units 2, 5, 6, 7, and 11. It may also be beneficial to take the time to write a positive note to each student.